Saint Columban College
College of Computing Studies
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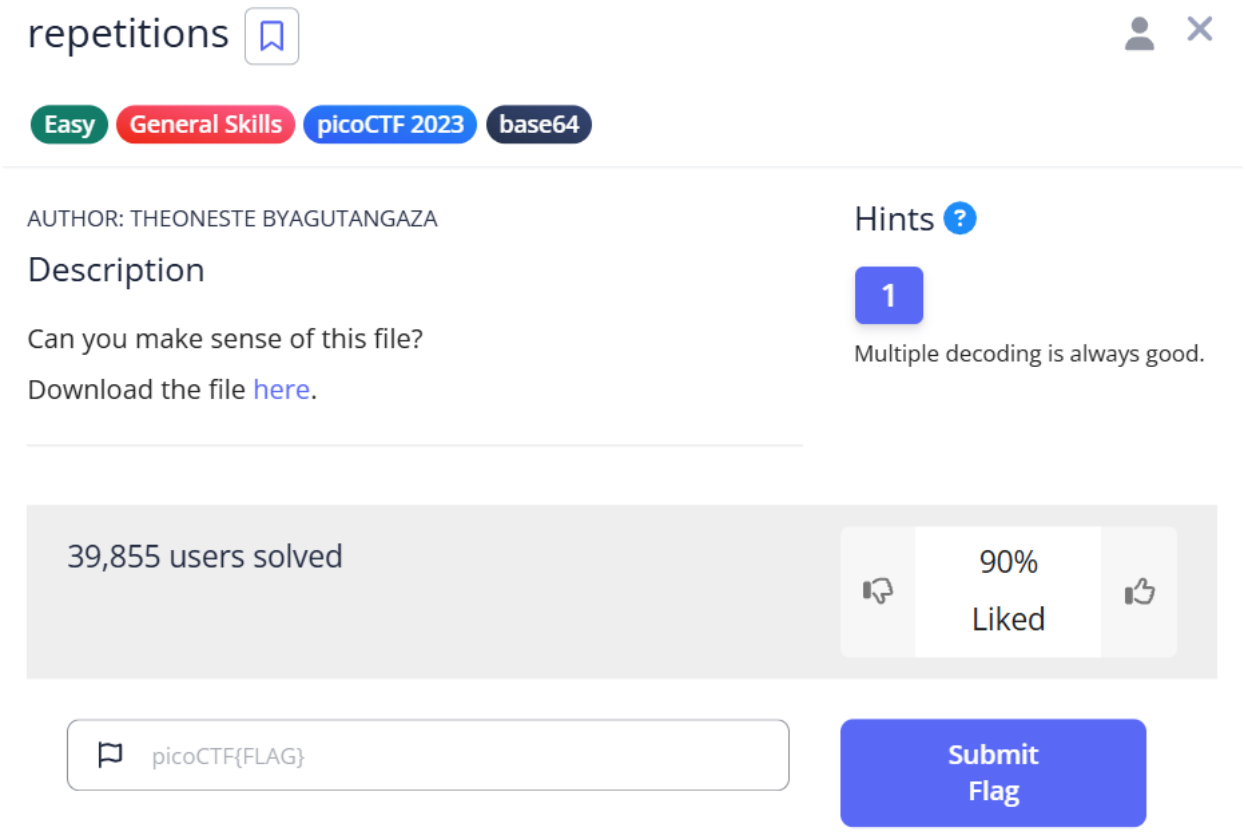
Repetitions



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Challenge: Repetitions

Category: General Skills

Date: 01/16/26

I. Objective

To retrieve the hidden flag by repeatedly decoding the contents of the provided file until readable text is obtained.

II. Background

Base64 is a common **encoding scheme** used to represent binary data in a text format. It is widely used in programming, file transfers, and CTF challenges because it safely represents data using readable characters.

In CTFs, Base64 is often **nested multiple times**, meaning the decoded output is still Base64-encoded and must be decoded again. This challenge demonstrates that concept clearly.

III. Tool Used

- **Oracle VirtualBox (Kali Linux)** – Used as the Linux environment
- **Linux Terminal** – Used to execute decoding commands
- **base64 utility** – Built-in Linux tool for decoding Base64 text

IV. Methodology

1. Downloaded the provided file and navigated to its directory.
2. Displayed the file contents using the cat command.
3. Observed that the text consisted of characters commonly associated with Base64 encoding.
4. Used **Ctrl + Shift + C** to copy and **Ctrl + Shift + V** to paste within the Linux terminal for efficiency.

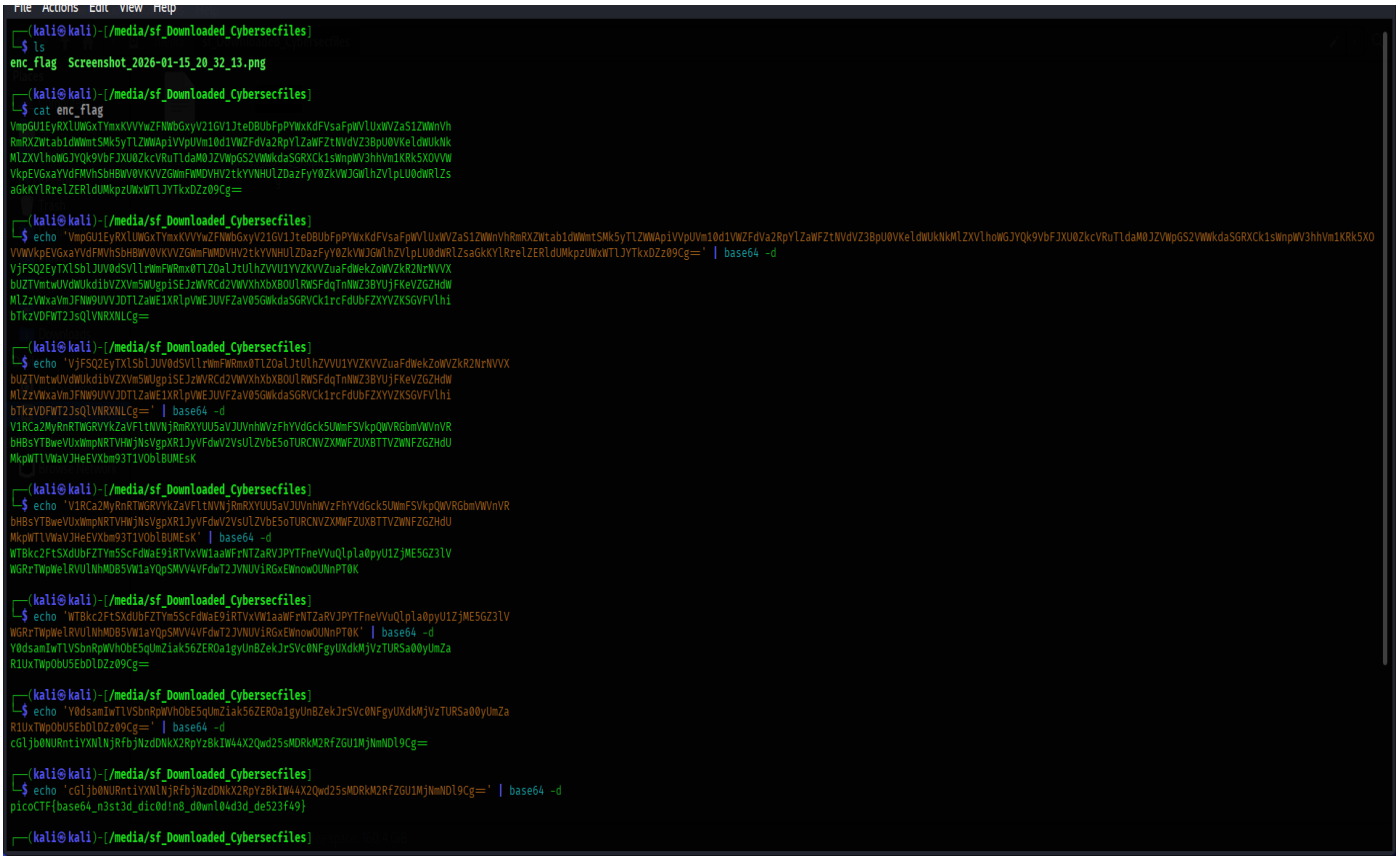
5. Decoded the file using the command:

   echo "<encoded_text>" | base64 -d

6. Noticed that the output was **still Base64-encoded**, so the same command was repeated.
7. Continued decoding multiple times—similar to opening a **Russian nesting doll (Matryoshka)** where each layer contains another encoded layer.
8. Repeated this process until the final output revealed a readable picoCTF flag.

V. Result

picoCTF{base64_n3st3d_dic0d!n8_d0wnl04d3d_de523f49}



VI. Explanation

Base64 is an encoding method that converts data into a set of 64 readable characters:

- Uppercase letters (A–Z)
- Lowercase letters (a–z)
- Numbers (0–9)
- Symbols (+ and /)
- Often ends with padding characters like = or ==

You can usually **identify Base64** when:

- The text contains only letters, numbers, +, /, and =
- The output looks “random” but decodes cleanly without errors
- Decoding produces another similarly formatted string

This challenge relies on **repetition** rather than complexity. Each decoded layer reveals another Base64 string, requiring the same decoding process again and again. The challenge teaches persistence and pattern recognition rather than advanced cryptography.

VII. Conclusion

The **Repetitions** challenge demonstrates how Base64 encoding can be layered multiple times to obscure data. By recognizing Base64 patterns and repeatedly applying the correct decoding method, the **hidden flag can be successfully uncovered.** This challenge reinforces the importance of patience, observation, and familiarity with basic encoding techniques in cybersecurity challenges.

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